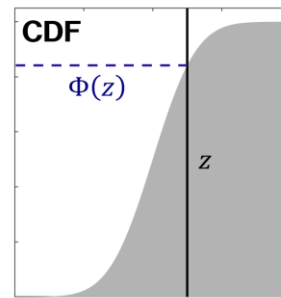
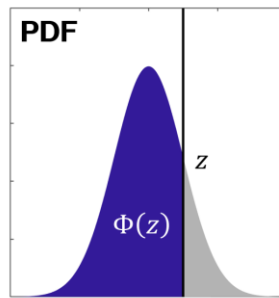


STATISTICAL TABLES 2024/25

Table 1: Cumulative Distribution Function for Standard Normal Distribution, $\Phi(z)$

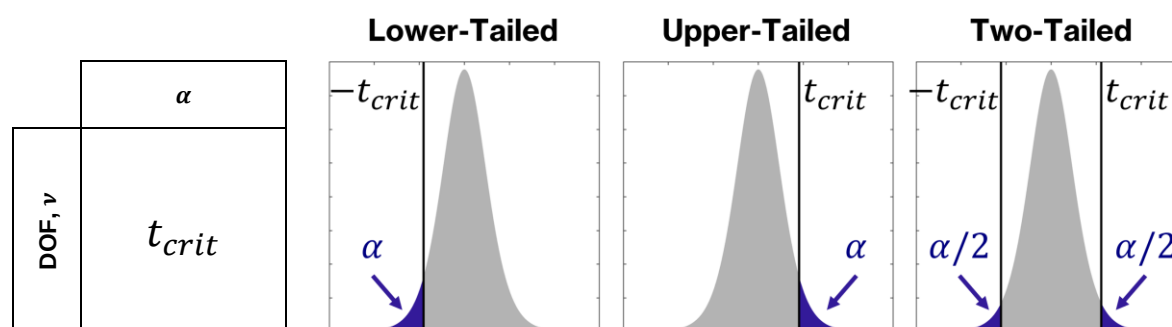
z (dp 2)	
z (dp 1)	$\Phi(z)$



Positive values only given. For negative values, use $\Phi(-z) = 1 - \Phi(z)$

Note e.g. $z_{1-\alpha/2}$ is equivalent to $\Phi(z) = 1 - \alpha/2$ or equivalently $z = \Phi^{-1}(1 - \alpha/2)$

		Second decimal place of z									
		0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
First decimal place of z	0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
	0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
	0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
	0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
	0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
	0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
	0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
	0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
	0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
	0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
	1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
	1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
	1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
	1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
	1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
	1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
	1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
	1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
	1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
	1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
	2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
	2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
	2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
	2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
	2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
	2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
	2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
	2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
	2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
	2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
	3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
	3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
	3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
	3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
	3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Table 2: Critical values of the t -distribution

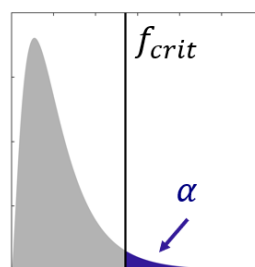
For 1-tailed, $t_{crit} = t_{1-\alpha, \nu} = F^{-1}(1 - \alpha, \nu)$

For 2-tailed, $t_{crit} = t_{1-\alpha/2, \nu} = F^{-1}(1 - \alpha/2, \nu)$

		α							
		1-tailed	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
		2-tailed	0.2	0.1	0.05	0.02	0.01	0.002	0.001
Degrees of Freedom, ν	1	3.078	6.314	12.71	31.82	63.66	318.3	636.6	
	2	1.886	2.920	4.303	6.965	9.925	22.33	31.60	
	3	1.638	2.353	3.182	4.541	5.841	10.22	12.92	
	4	1.533	2.132	2.776	3.747	4.604	7.173	8.610	
	5	1.476	2.015	2.571	3.365	4.032	5.893	6.869	
	6	1.440	1.943	2.447	3.143	3.707	5.208	5.959	
	7	1.415	1.895	2.365	2.998	3.499	4.785	5.408	
	8	1.397	1.860	2.306	2.896	3.355	4.501	5.041	
	9	1.383	1.833	2.262	2.821	3.250	4.297	4.781	
	10	1.372	1.812	2.228	2.764	3.169	4.144	4.587	
	11	1.363	1.796	2.201	2.718	3.106	4.025	4.437	
	12	1.356	1.782	2.179	2.681	3.055	3.930	4.318	
	13	1.350	1.771	2.160	2.650	3.012	3.852	4.221	
	14	1.345	1.761	2.145	2.624	2.977	3.787	4.140	
	15	1.341	1.753	2.131	2.602	2.947	3.733	4.073	
	16	1.337	1.746	2.120	2.583	2.921	3.686	4.015	
	17	1.333	1.740	2.110	2.567	2.898	3.646	3.965	
	18	1.330	1.734	2.101	2.552	2.878	3.610	3.922	
	19	1.328	1.729	2.093	2.539	2.861	3.579	3.883	
	20	1.325	1.725	2.086	2.528	2.845	3.552	3.850	
	40	1.303	1.684	2.021	2.423	2.704	3.307	3.551	
	60	1.296	1.671	2.000	2.390	2.660	3.232	3.460	
	120	1.289	1.658	1.980	2.358	2.617	3.160	3.373	
	∞	1.282	1.645	1.960	2.326	2.576	3.090	3.291	

Table 3: Selected Critical values of the F-distribution for $\alpha = 0.05$

	ν_1
ν_2	F_{crit}



$$F_{crit} = f_{1-\alpha, \nu_1, \nu_2} = F^{-1}(1 - \alpha, \nu_1, \nu_2)$$

		ν_1																
		1	2	3	4	5	6	7	8	9	10	12	15	20	40	60	120	Inf
ν_2	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.72	5.69	5.66	5.63
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.46	4.43	4.40	4.36
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.77	3.74	3.70	3.67
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.34	3.30	3.27	3.23
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.04	3.01	2.97	2.93
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.83	2.79	2.75	2.71
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.66	2.62	2.58	2.54
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.53	2.49	2.45	2.40
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.43	2.38	2.34	2.30
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.34	2.30	2.25	2.21
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.27	2.22	2.18	2.13
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.20	2.16	2.11	2.07
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.15	2.11	2.06	2.01
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.10	2.06	2.01	1.96
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.06	2.02	1.97	1.92
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.03	1.98	1.93	1.88
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	1.99	1.95	1.90	1.84
	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.69	1.64	1.58	1.51
	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.59	1.53	1.47	1.39
	120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.50	1.43	1.35	1.25
	Inf	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.39	1.32	1.22	1.00

Table 4: Selected Critical values of the χ^2 -distribution

α		0.050	0.010	0.001
ν	-	-	-	-
	1	3.841	6.635	10.828
	2	5.991	9.210	13.816
	3	7.815	11.345	16.266
	4	9.488	13.277	18.467
	5	11.070	15.086	20.515
	6	12.592	16.812	22.458
	7	14.067	18.475	24.322
	8	15.507	20.090	26.124
	9	16.919	21.666	27.877
	10	18.307	23.209	29.588

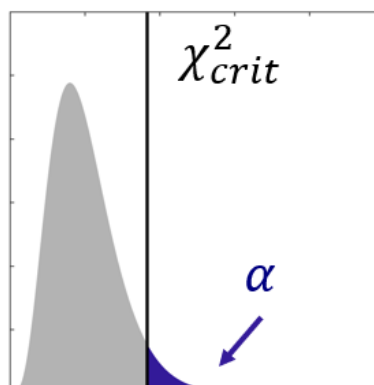


Table 5: Critical values for the Sign-Rank Test

n	One-Sided $\alpha = 0.01$ Two-Sided $\alpha = 0.02$	One-Sided $\alpha = 0.025$ Two-Sided $\alpha = 0.05$	One-Sided $\alpha = 0.05$ Two-Sided $\alpha = 0.1$
5			1
6		1	2
7	0	2	4
8	2	4	6
9	3	6	8
10	5	8	11
11	7	11	14
12	10	14	17
13	13	17	21
14	16	21	26
15	20	25	30
16	24	30	36
17	28	35	41
18	33	40	47
19	38	46	54
20	43	52	60
21	49	59	68
22	56	66	75
23	62	73	83
24	69	81	92
25	77	90	101
26	85	98	110
27	93	107	120
28	102	117	130
29	111	127	141
30	120	137	152

Table 6: Critical values of the U-statistic for the Rank-Sum test

Note that the Wilcoxon rank-sum test statistic is related to the u -statistic according to

$$u_i = w_i - \frac{n_i(n_i + 1)}{2}$$

One-Tailed Test at $\alpha = 0.025$ or Two-Tailed Test at $\alpha = 0.05$																	
n_1	n_2																
	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1																	
2					0	0	0	0	1	1	1	1	1	2	2	2	2
3		0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8
4	0	1	2	3	4	4	5	6	7	8	9	10	11	11	12	13	13
5		2	3	5	6	7	8	9	11	12	13	14	15	17	18	19	20
6			5	6	8	10	11	13	14	16	17	19	21	22	24	25	27
7				8	10	12	14	16	18	20	22	24	26	28	30	32	34
8					13	15	17	19	22	24	26	29	31	34	36	38	41
9						17	20	23	26	28	31	34	37	39	42	45	48
10							23	26	29	33	36	39	42	45	48	52	55
11								30	33	37	40	44	47	51	55	58	62
12									37	41	45	49	53	57	61	65	69
13										45	50	54	59	63	67	72	76
14											55	59	64	67	74	78	83
15												64	70	75	80	85	90
16													75	81	86	92	98
17														87	93	99	105
18															99	106	112
19																113	119
20																	127

One-Tailed Test at $\alpha = 0.05$ or Two-Tailed Test at $\alpha = 0.1$																	
n_1	n_2																
	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	20
1																	0
2			0	0	0	1	1	1	1	2	2	3	3	3	3	4	0
3	0	0	1	2	2	3	4	4	5	5	6	7	7	8	9	9	4
4		1	2	3	4	5	6	7	8	9	10	11	12	14	15	16	11
5			4	5	6	8	9	11	12	13	15	16	18	19	20	22	18
6				7	8	10	12	14	16	17	19	21	23	25	26	28	25
7					11	13	15	17	19	21	24	26	28	30	33	35	32
8						15	18	20	23	26	28	31	33	36	39	41	39
9							21	24	27	30	33	36	39	42	45	48	47
10								27	31	34	37	41	44	48	51	55	54
11									34	38	42	46	50	54	57	61	62
12										42	47	51	55	60	64	68	69
13											51	56	61	65	70	75	77
14												61	66	71	77	82	84
15													72	77	83	88	92
16														83	89	95	100
17															96	102	107
18																109	115
19																	123
20																	138